

☑ Also known as React.js/ React.JS

✳ **Free & Open-Source JavaScript Library for building user interfaces (UIs).**

➡ React is **Created by Meta (Facebook)** & is now maintained by Meta & community of developers.

- HTML/CSS/JS code refreshes the whole page unexpectedly upon interaction (like clicking a button, submitting a form, etc.)

React Features:

- React is used to build *SPAs (Single Page Applications)*
- Creates *reusable UI components*
- Develops dynamic, fast & scalable front end applications.
- ReactJS uses a Virtual DOM mechanism to efficiently update the HTML DOM.
- Faster Navigation (No full page reloads)
- Allows writing HTML-like syntax inside JavaScript through JSX.
- Data flows in a single direction (parent → child).
- With React Native, the same concepts can be used to build mobile apps for Android and iOS.
- Huge ecosystem of libraries, tools, and community support.

🔑 ReactJS uses a *Virtual DOM* mechanism to efficiently update the *HTML DOM*.

Instead of reloading the entire DOM, the virtual DOM updates only the specific elements that have changed, resulting in faster performance.

🏢 React follows a *Component-Driven Architecture*

The user interface is built by composing independent, reusable pieces called components. Each component manages its own logic and rendering, making development more modular, scalable, and easier to maintain.

💡 **A component is a reusable piece of code that represents a specific part of the user interface (UI).**

React Element : An object that describes the properties of an actual DOM node which will be created by React.

✳ It contains information (like **type**, **props**, and **children**) about the actual DOM node that React will create and manage.

Simple Structure of a React Elements:

```
{
  type: "button",
  props: {
    className: "btn",
```

```
    children: "Click Me"  
  }  
}
```

▣ Explanation:

- **type** → The type of DOM element (e.g., "div", "button", etc.)
- **props** → The attributes or properties passed to the element
- **children** → The inner content or nested elements (optional)